UI/UX DESIGN SPECIFICATION REPORT

Kemet Hieroglyphic Studio: Interface Architecture, Component Specs & Design System

Track: DEPI Round 5 (Machine Learning)

Brand & Theme

1. Design System & Visual Identity Monumental Ancient Meets Modern AI

Combines classical Egyptian heritage aesthetics with clean, responsive SaaS usability.

Color Palette & Tokens

- Deep Navy (#0A2947): Navbar, headers & dark badges.
- Warm Papyrus (#F3E4C9): Main app canvas & card backgrounds.
- Terracotta (#8B5E3C): Primary buttons, active tabs & action calls.
- Muted Limestone (#D3D4C0): Accent pills & subtle tags.
- Light Papyrus (#FBF5EB): Input fields & conversational chat bubbles.

Typography & Navigation Model

Display Font:
Classical Serif for headings & hieroglyphic labels.

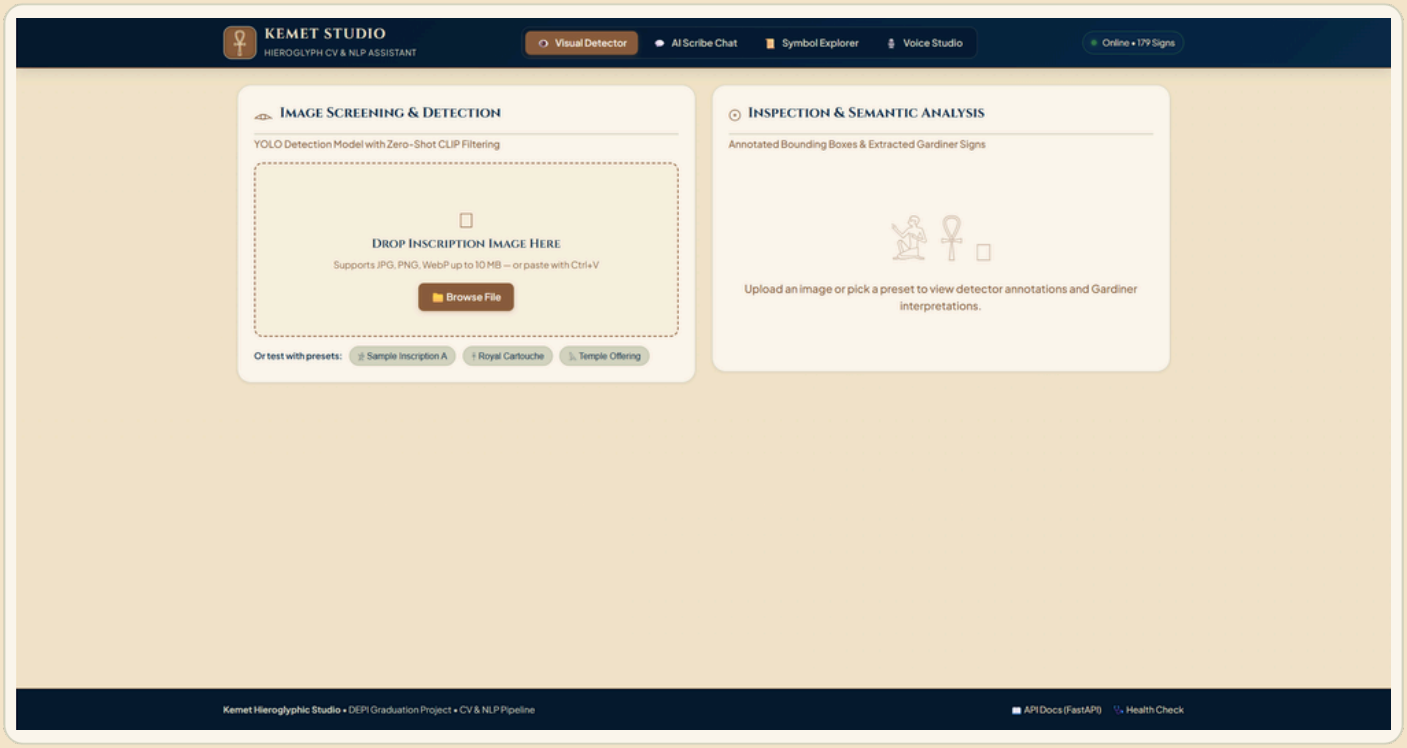
Body Font:
Highly readable Sans-Serif for descriptions & chat.

Sticky Header:
Brand Ankh mark, 4 tab buttons & status badge (Online • 179 Signs).

Persistent Footer:
FastAPI status link & service health check.

2. Core Interface Modules & Interaction Design

Module 1: Visual Detector (Image Screening & Detection)



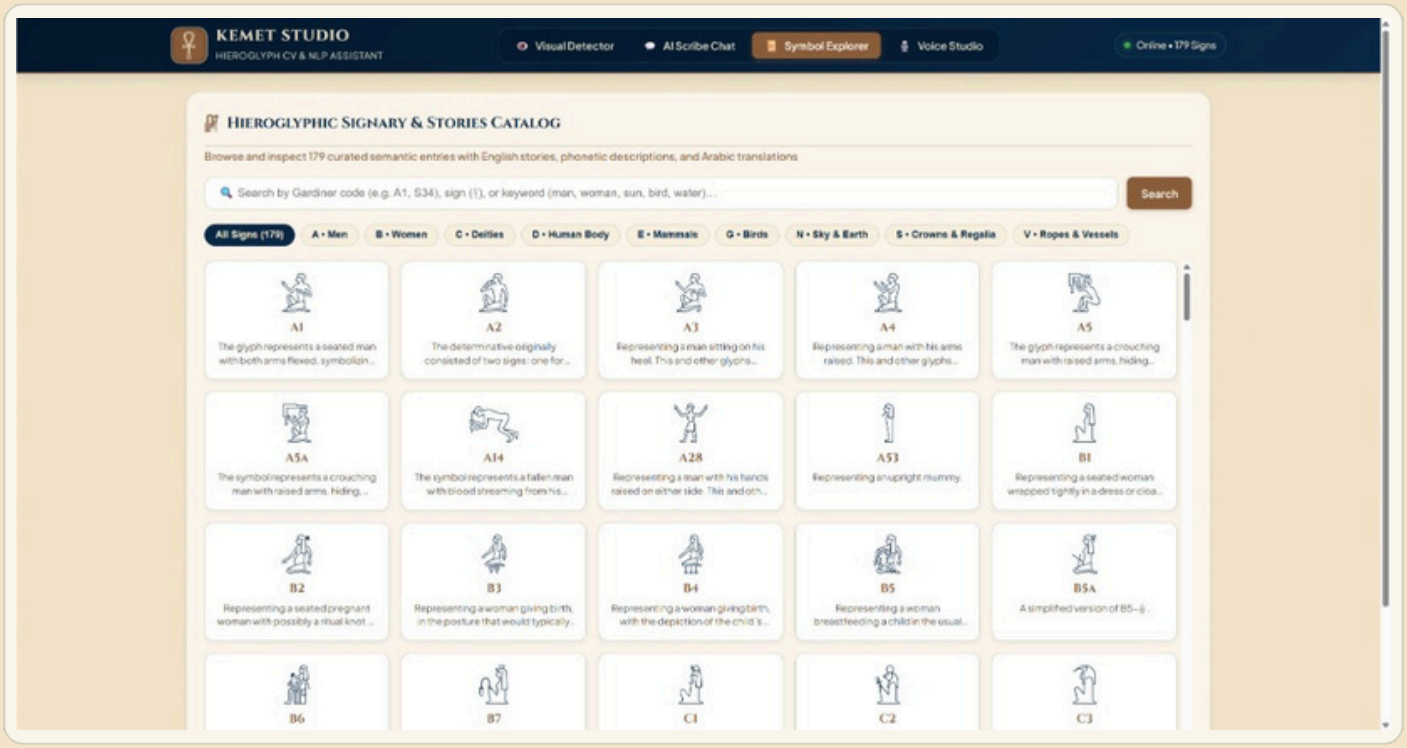
Upload Workflow:
Supports drag-and-drop, file browsing, and clipboard paste (Ctrl+V) up to 10MB.

Preset Exploration:
1-click test cards (Sample Inscription A, Royal Cartouche, Temple Offering) for instant onboarding.

Visual Feedback:
Displays model bounding boxes, detection confidence scores, and extracted Gardiner codes side-by-side with the uploaded artifact.

2. Core Interface Modules & Interaction Design

Module 2: Symbol Explorer (Hieroglyphic Signary & Stories)



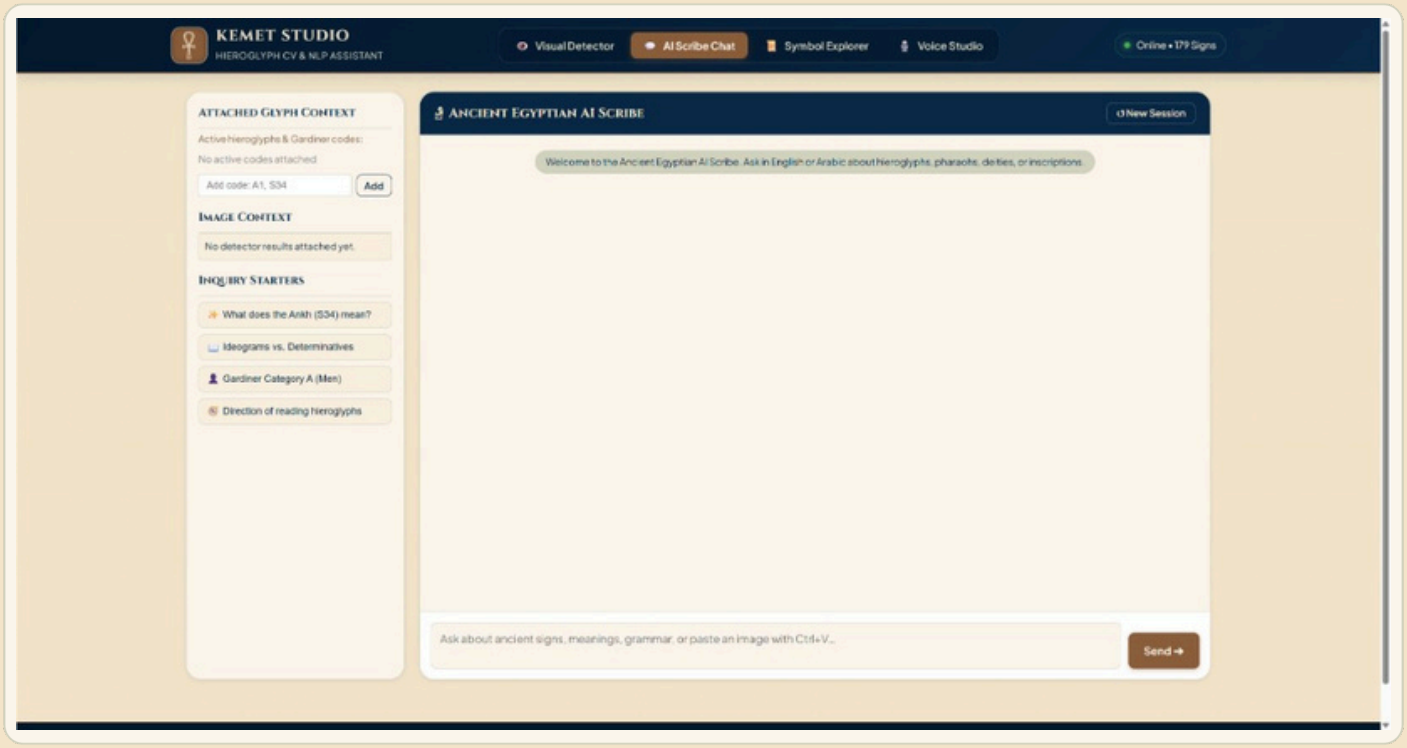
Search & Filtering:
Real-time search across Gardiner codes, Unicode glyphs, and natural keywords (sun, water, man).

Category Navigation:
Filter pills for 10 classifications (All 179 Signs, Men, Women, Deities, Body, Birds, etc.).

Visual Card Grid:
Responsive cards pairing high-contrast glyph line art with code labels and concise cultural descriptions.

2. Core Interface Modules & Interaction Design

Module 3: AI Scribe Chat (Contextual Conversational Q&A;)



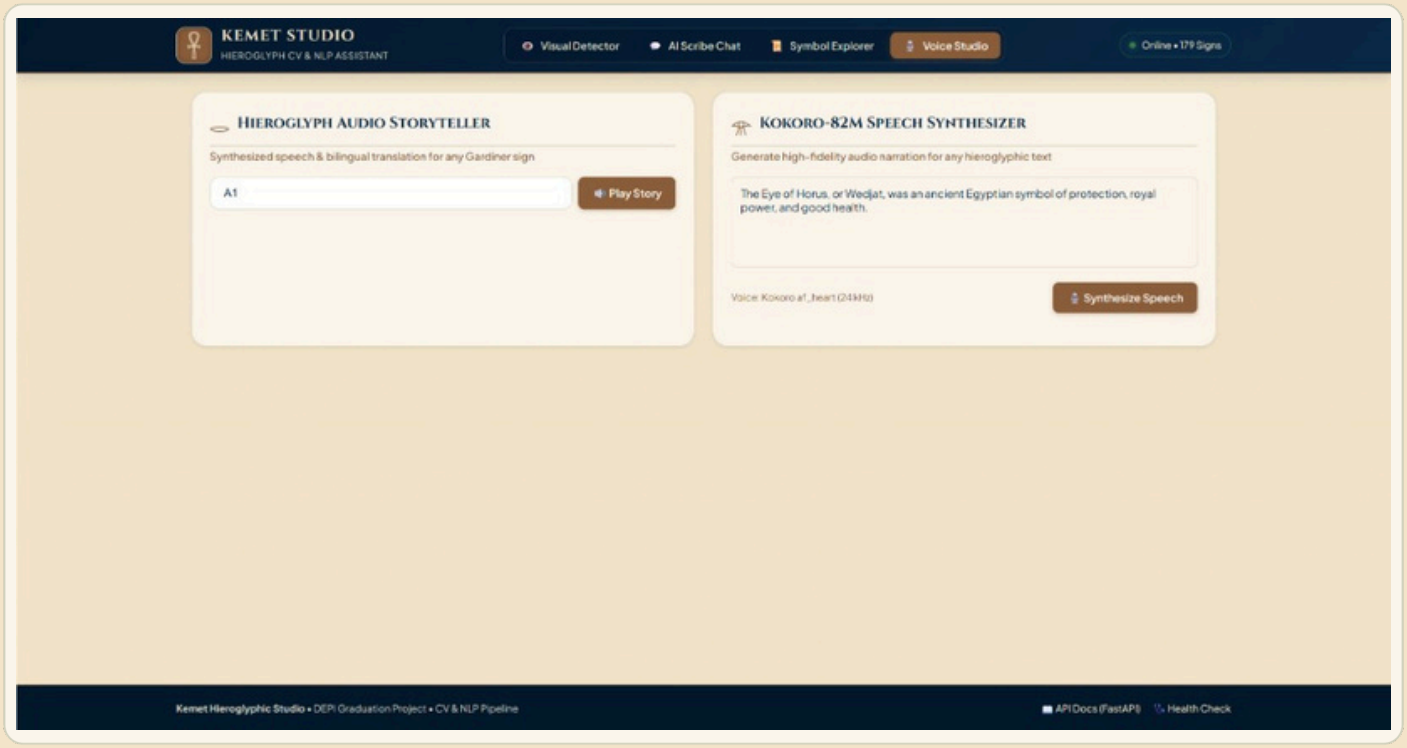
Context Sidebar:
Tracks attached active glyphs, allows manual Gardiner code entry, and retains detection state.

Inquiry Starters:
Suggested prompt pills (e.g. Ankh meaning, Ideograms vs. Determinatives, Reading direction).

Authentic Typography:
Renders verified Unicode Egyptian hieroglyphs directly inside conversation bubbles in English and Arabic.

2. Core Interface Modules & Interaction Design

Module 4: Voice Studio (Audio Storytelling & Speech Synthesis)

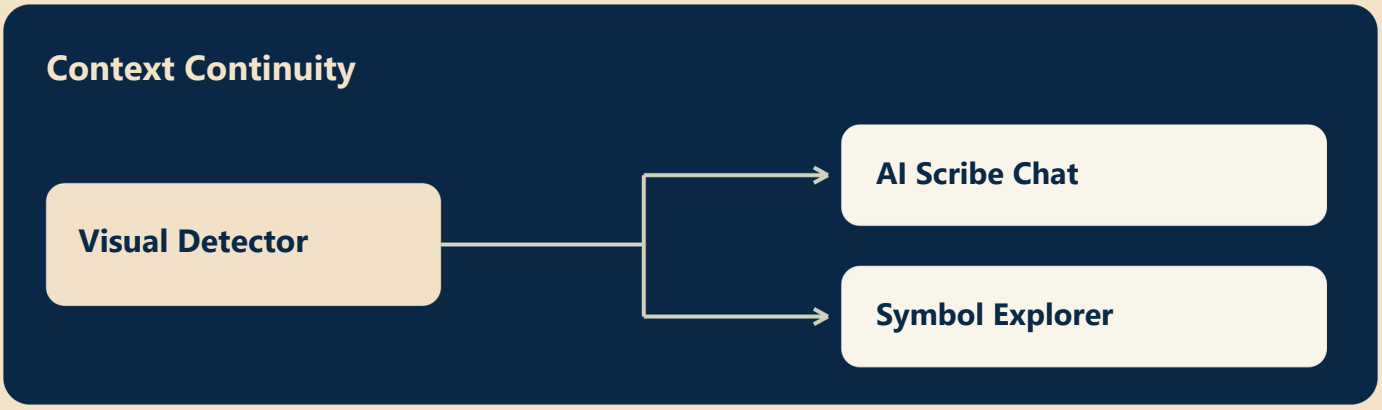


Hieroglyph Storyteller:
Dedicated card to play synthesized spoken stories and translations for any Gardiner sign.

Kokoro-82M Synthesizer:
Generates high-fidelity 24 kHz spoken audio narration (af_heart voice) for custom text.

Accessible Controls:
Instant audio playback with browser-compatible HTML5 audio player and visual audio wave indicator.

3. Cross-Module Interaction & Accessibility Guidelines



Interaction Dimension

UI/UX Design Decision & User Experience Guarantee

Context Continuity

Detected signs from Visual Detector automatically sync into the AI Scribe Chat context and Symbol Explorer detail panel without page reloads.

Transparency & Trust

Model confidence levels are framed as suggestions; clear disclaimers state that AI responses assist exploration without replacing formal scholarship.

Screen Responsiveness

Viewports maintain independent scrollable sidebars, preventing overlap on laptop screens; layout adapts fluidly between desktop and tablet viewports.